

IMO (55th) – 2014

First Day

1. Let $a_0 < a_1 < \dots$ be a finite sequence of positive integers. Prove that there exist a unique integer $n \geq 1$ such that $a_n \leq \frac{a_0 + a_1 + \dots + a_n}{n} \leq a_{n+1}$.
2. Let $n \geq 2$ be an integer. Consider an $n \times n$ chessboard consisting of n^2 unit squares. A configuration of n rooks on this board is peaceful if every row and every column contains exactly one rook. Find the greatest positive integer k such that for each powerful configuration of n rooks, there is $k \times k$ square which does not contain a rook on any of its k^2 unit squares.
3. Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and $\angle CHS - \angle CSB = 90^\circ$, $\angle THC - \angle DTC = 90^\circ$. Prove that line BD is tangent to the circumcircle of triangle TSH .

Second Day

4. A point P and Q lie on side BC of acute-angled triangle ABC so that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Points M and N lie on lines AP and AQ respectively, such that P is the midpoint of AM and Q is the midpoint of AN . Prove that lines BM and CN intersect on the circumcircle of triangle ABC .
5. For each positive integer n , the Bank of Cape town issues coins of denomination $\frac{1}{n}$. Given a finite collection of such coins (of not necessarily different denominations) with total value at most $99 + \frac{1}{2}$. Prove that it is possible to split this collection into 100 or fewer groups, such that each group has total value at most 1.
6. A set of lines in the plane is in general position if no two are parallel and no three pass through the same point. A set of lines in general position cuts the plane into regions, some of which have finite area; we call these its finite regions. Prove that for all sufficiently large n , in any set of n lines in general position it is possible to colour at least $\sqrt{n}$ of the lines blue in such a way that none of its finite regions has a completely blue boundary.

Note: Results with $\sqrt{n}$ replaced by $c\sqrt{n}$ will be awarded points depending on the value of the constant c .
